

THE MULTIPLE EFFECT EVAPORATOR IN THE RAW SUGAR FACTORY

by

H. S. Birkett
F. C. Schaffer & Associates, Inc.

For Presentation at the Annual Meeting
American Society of Sugarcane Technologists
Louisiana Division, February 10, 1978

This paper presents thirteen evaporator schemes, worked out in detail, each of which performs the same duty. These evaporator schemes are compared with respect to exhaust requirements, heating surface requirements, and condenser loads (injection water requirements). The effect of vapor bleeding (vapor robbing) can be deduced by comparing the evaporator schemes. This paper covers triple effect, quadruple effect, and quintuple effect evaporators in conjunction with several vapor bleeding alternatives, and the addition of pre-evaporators.

The design and operation of the evaporator station is central to the energy balance of a sugar factory. The selection of the appropriate evaporator station is of paramount importance in designing a sugar factory to operate satisfactorily under given conditions, and within fixed constraints.

In this paper the rigorous heat balance method of calculation has been adopted. It is felt that the simplified and short-cut methods reported in the literature from time to time, are not sufficiently accurate for a definitive comparison of several schemes. Further, in the simplified methods, it is not generally possible to predict with any degree of accuracy

the actual operating pressures that will result throughout the evaporator set, and the accurate determination of these intermediate pressures becomes important in sizing of the units (heaters and pans) that operate off first and second vapor.

In Louisiana, vapor bleeding has been the exception rather than the rule. Further, triple effect evaporators have been the overwhelming favorite for evaporation. The present energy situation has highlighted the gross energy inefficiency of Louisiana sugar factories. Louisiana factories will have to change their evaporation stations (along with other changes in the factory), if they are to realize the achievable goal of energy self-sufficiency.

This paper considers only those changes at the evaporator and heater stations that will be required for fuel self-sufficiency.

This paper has not considered the use of thermo-compressors (or turbo-compressors) as a means of decreasing the evaporator's exhaust requirements, since their suitability can only be meaningfully evaluated in the context of the factory's total energy balance due to their requirement for live steam.

DISCUSSION OF RESULTS

Heating Surface:

The evaporator station heating surface, as well as the total heating surface for the evaporator station and heaters is primarily determined by the number of effects. The average total heating surface for the evaporator and heaters per 100 tons of juice per hour is as follows:

Triple Effect	17,700 sq. ft.
Quadruple Effect	22,700 sq. ft.
Quintuple Effect	32,300 sq. ft.

The addition of a pre-evaporator, or the utilization of evaporator bleeds has very little effect on the total heating surface requirements, except for the case of the quintuple effect evaporator where the boiling point elevations begin to significantly affect the net overall temperature drop.

At the present time, both evaporator and heater heating surface costs between \$9 - 12 per sq. ft. not including installation costs. Installation costs often run 50% to 100% of the cost of the equipment. The total cost of each arrangement can be considered as being proportional to the total heating surface required for both evaporator and heaters.

The heating surface (and hence cost) for the various evaporator schemes is shown in Table 1. The total heating surface relative to that of a straight triple effect for the various evaporator cases is also given in Table 1.

Steam Requirement:

The primary reason for the use of multiple effect evaporators results from the reduction in exhaust steam usage that they afford. In general, the greater the number of effects employed, the greater the quantity of vapors bled, and the farther down the set that vapor is bled, the greater is the exhaust steam saving.

Table 1 tabulates the total (evaporators, pans, and heaters) boiling house exhaust steam requirements for each evaporator case. Also tabulated in Table 1 is the relative exhaust steam required for each case as a percentage of that required by a straight triple effect.

It will be noted that an evaporator preceded by a pre-evaporator functions fundamentally as the corresponding evaporator scheme without a pre-evaporator in which

the pre-evaporator vapor load is supplied by the first effect. Thus, Case 3 (triple with first vapor bled to all pans and heaters) and Case 5 (pre-evaporator supplying vapor to all pans and heaters followed by a triple effect) are virtually identical in heating surface requirement, exhaust requirement, and injection water requirement. The corresponding cases for a quadruple effect evaporator, Cases 8 and 10, show the same similarity. In other words, the pre-evaporator functions as an extension of the first effect.

It can be seen from Table 1 that the double robbed quadruple effect evaporator (Case 9) requires only 70.8% of the exhaust steam required by a triple effect evaporator. The use of a triple robbed quintuple effect evaporator (Case 13) can reduce the exhaust steam requirement to only 63% of that of a straight triple effect evaporator.

Under Louisiana conditions where the ash % bagasse is very high (up to 20%), and boiler efficiencies are relatively low (about 50%), the exhaust steam available from bagasse combustion generally cannot support any evaporator arrangement less efficient than a robbed quadruple effect evaporator. Since the overwhelming majority of Louisiana factories employ triple effect evaporators, the result is that considerable quantities of auxiliary fuel (primarily natural gas) are consumed. In 1977, the average Louisiana factory used 0.6 MCF of gas per ton of cane. By comparison, some Louisiana factories that sold all their bagasse for by-product usage required about 1.8 MCF of gas per ton of cane even though these factories had triple effect evaporators. In other words, the average bagasse burning Louisiana factory is only supplying 65% of its total steam requirements from bagasse. This compares very unfavorably with other cane producing areas where factories are entirely fuel self-sufficient on bagasse.

It should be noted, however, that the installation of more efficient evaporator schemes by themselves in the Louisiana factories will not necessarily result in lower auxiliary fuel usage, unless the live steam requirement of factory prime movers can also be reduced. However, changes to more efficient evaporator schemes are a prerequisite for energy self-sufficiency.

One reason for the reluctance of some factories to extensively employ vapor heating is the reduced pressure that occurs when the evaporators are clean. The vapor conditions shown in the accompanying sketches only apply when the coefficients assumed are also the actual coefficients. In reality, when the evaporator has been just cleaned the applicable heat transfer coefficients are appreciably better. The result is that the differential temperature (and hence pressure) in the effects declines. The result is that all pressures except that in the body of the last effect decline. This can have adverse effects on units operating on vapor unless precautions are taken to maintain the vapor pressure of robbed bodies. The most suitable method of maintaining adequate pressure in the robbed effects is to place a control valve in the vapor line between the last robbed effect and the following effects. This permits the maintenance of adequate pressure in the robbed effects while not exceeding the required evaporation rate in the evaporator. In extreme cases a control valve might also be required between each robbed effect.

Pre-Evaporators:

As noted earlier, an evaporator set with a pre-evaporator achieves essentially the same steam economy as an evaporator set without a pre-evaporator provided that the set without the pre-evaporator has a first effect heating surface equal to that of the sum of the pre-evaporator and first effect, and where the pre-evaporator load is absorbed by the first evaporator effect.

Pre-evaporator schemes, however, have operating advantages over robbed first vessels. Pan loads, which fluctuate widely due to their batch operation are best handled by pre-evaporators since the pre-evaporator can respond rapidly to meet varying vapor loads without affecting the boiling conditions in the evaporator bodies. More stable and flexible evaporator performance and control is possible when the vapor generation function of the first effect is performed in a separate pre-evaporator, because the vapor pressure of the pre-evaporator is independent of that of the first effect.

Injection Water Requirement:

The evaporator's last effect vapors to be condensed, together with the injection water required (based on 100°F inlet injection water temperature and 120°F condenser water temperature) are tabulated in Table 1. The evaporator's and total (evaporator and pans) injection water requirement relative to that of a straight triple effect is given in Table 1. It is interesting to note that the double robbed quadruple effect evaporator (Case 9) requires only 37.5% of the water requirements for a straight triple effect evaporator, and that the corresponding total (evaporator plus pans) injection water requirement is only 55.9% of that of the straight triple effect case.

The reduced injection water requirements of the vapor robbed quadruple effect evaporators can be an important consideration in reducing entrainment and cooling pond requirements.

Factory Condensate Balance:

Most of the water boiled out of the juice by the evaporators is recovered as condensate from the second, third, . . . and last effect's with only the evaporation from the last effect being lost (to the condensers). Evaporation schemes with small condenser loads provide more condensate insuring a favorable condensate balance.

Entrainment:

The evaporation rates achieved by triple, quadruple, and quintuple effect evaporators are of the order of 11, 8, and 6 lbs./hr.-ft.² respectively. Entrainment from an effect is proportional to both the evaporation occurring in the effect, and the square of the vapor velocity.

The percentage of entrainment increases as the effect number increases because of the increased vapor velocities resulting from the increased specific volume of the vapor at the lower pressures. The evaporator schemes with low evaporation rates and/or reduced last effect vapor loads result in reduced entrainment losses. Thus the entrainment from the last effect of the triple-robbed quintuple effect evaporator (Case 13) is only 4.2% of that of the straight triple effect evaporator (Case 1).

THE RIGOROUS EVAPORATOR HEAT BALANCE

The rigorous heat balance entails that at least the following conditions be met:

- a) The heat required for juice heating in the first effect (and/or pre-evaporator) should be taken into account.
- b) The variation of the specific heat of the juice should be taken into account in calculating the flash in each of the evaporator bodies.
- c) The operating pressures in each body should be determined by consideration of the heat transfer coefficients of each body which set the temperature difference required in each body.
- d) Allowance should be made for both the boiling point elevation due to concentration and the hydrostatic head loss in each body.

A TRIAL AND ERROR METHOD FOR THE DETERMINATION OF THE RIGOROUS EVAPORATOR HEAT BALANCE

The method used in calculating the several heat balances in this paper is as follows:

- 1) Assume pressures in each vapor space of each evaporator body. This first assumption may be made assuming equal pressure drop throughout the evaporator set.
- 2) Obtain the corresponding saturated vapor temperature and latent heats of evaporation.
- 3) Estimate the evaporation in each body by a short cut method, and calculate the Brix into and out of each effect.
- 4) Calculate the boiling point elevation in each effect due to the concentration and the hydrostatic head effect.
- 5) Assume a value for the exhaust required (this can be obtained for the first trial by a short cut method), and using this value calculate the evaporation in, and juice out, of each effect.
- 6) The total evaporation obtained in Step 5 above is then compared with the required evaporation and a new estimate of the steam required is calculated and the operations of Step 5 and 6 are repeated until the exhaust assumption checks with the evaporation required to the nearest pound.
- 7) The specific heat, and boiling point elevations are then recalculated using the Brix obtained in the final calculation of Step 6, and Steps 5 and 6 repeated. The specific heat of sugar juices can be obtained from:

$$\text{Sp. Ht.} = -0.00549 (\text{Juice Brix}) + 0.9964$$

The specific heat thus obtained is in English units (Btu/Lb.-°F).

- 8) Step 7 is repeated until the assumptions agree with the calculated values.
- 9) The actual heat transfer coefficients are then calculated.

- 10) Expected coefficients are then calculated using the Dessin equation. The Dessin equation is:

$$\text{Overall heat transfer coefficient, } U = \frac{(100 - B)(T - 130) \lambda}{K}$$

where U is the heat transfer coefficient, Btu/Hr.-Sq. Ft.-°F

B is the Brix of the liquid in the effect

T is the temperature of the steam in the calandria, °F

λ is the latent heat of the vapor in the vapor space, Btu/Lb.

K is a constant, numerical value 20,000 for design purposes

- 11) The evaporator vapor conditions are then revised to give heat transfer coefficients that are in proportion to the design (Dessin) coefficients.
- 12) With these vapor conditions the entire sequence of steps are recalculated to get the new quantity of steam required.
- 13) The revision of the evaporator conditions is repeated until there is no change in conditions, and the design heat transfer coefficients for each body are proportional to the design (Dessin) heat transfer coefficients.
- 14) If the design is for a new evaporator, the areas are selected so as to make the actual and design coefficients equal. Additional heating surface may be provided if a greater margin of safety is required.

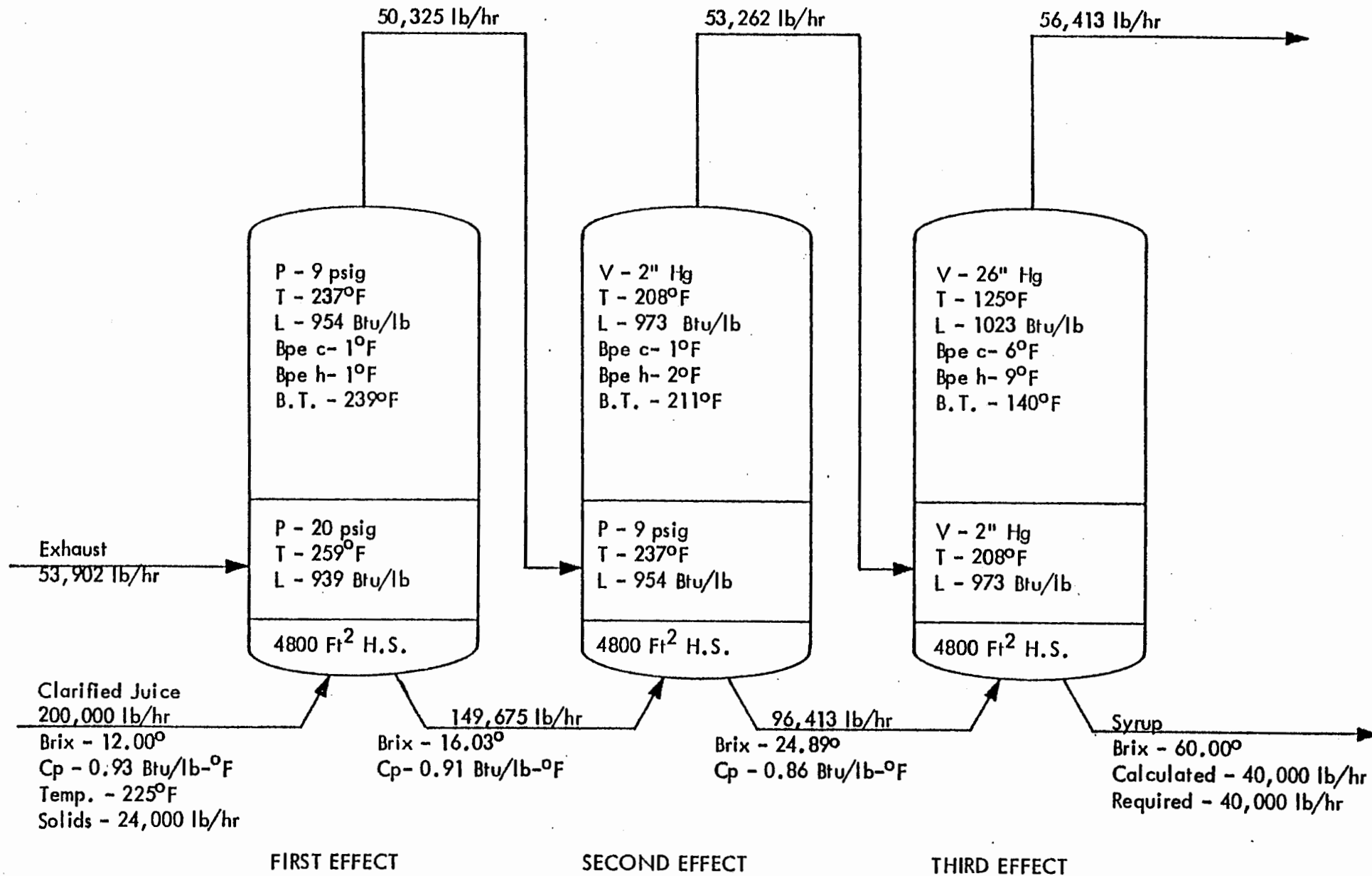
CASE 1

EVAPORATOR MATERIAL AND ENERGY BALANCE

	<u>Exhaust or Vapor, Lb/Hr</u>	<u>Heat Transferred Btu/Hr</u>	<u>Material Lb/Hr</u>	<u>°Brix</u>
<u>First Effect</u>				
Entering: (53,902) (939)	53,902	50,613,978	200,000	12.00
Less: (200,000) (0.93) (239-225)		2,604,000		
Available for Evaporation		<u>48,009,978</u>		
Evaporation: $48,009,978 \div 954$	50,325			
<u>Second Effect</u>				
Entering:	50,325	48,009,978	149,675	16.03
Plus: (149,675) (0.91) (239-211)		3,813,719		
Available for Evaporation		<u>51,823,697</u>		
Evaporation: $51,823,697 \div 973$	53,262			
<u>Third Effect</u>				
Entering:	53,262	51,823,697	96,413	24.89
Plus: (96,413) (0.86) (211-140)		5,886,978		
Available for Evaporation		<u>57,710,675</u>		
Evaporation: $57,710,675 \div 1023$	56,413			
Syrup to Pans		Calculated	40,000	60.00
		Required	40,000	60.00

CASE I

EVAPORATION CONDITIONS



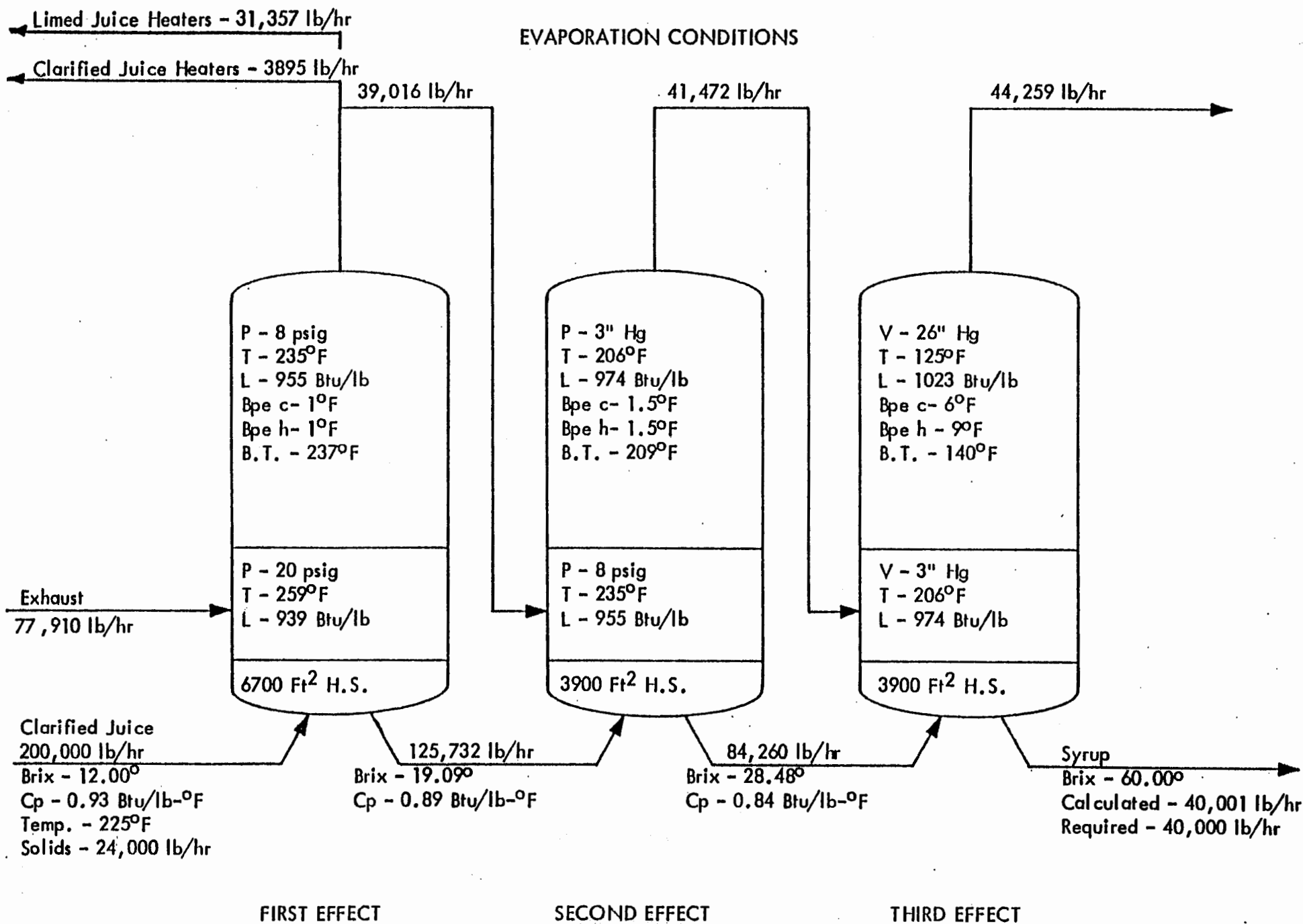
CASE 2

EVAPORATOR MATERIAL AND ENERGY BALANCE

	<u>Exhaust or Vapors, Lb/Hr</u>	<u>Heat Transferred, Btu/Hr</u>	<u>Material Lb/Hr</u>	<u>°Brix</u>
<u>First Effect</u>				
Entering: (77,910) (939)	77,910	73,157,490	200,000	12.00
Less: (200,000) (0.93) (237-225)		2,232,000		
Available for Evaporation		<u>70,925,490</u>		
Evaporation: $70,925,490 \div 955$	74,268			
Vapor to Lined Juice Heater ($29,946,000 \div 955$)	31,357			
Vapor to Clarified Juice Heater ($3,720,000 \div 955$)	3,895			
Vapor to Second Effect	39,016			
<u>Second Effect</u>				
Entering: (39,016) (955)	39,016	37,260,280	125,732	19.09
Plus: (125,732) (0.89) (237-209)		3,133,241		
Available for Evaporation		<u>40,393,521</u>		
Evaporation: $40,393,521 \div 974$	41,472			
<u>Third Effect</u>				
Entering:	41,472	40,393,521	84,260	28.48
Plus: (84,260) (0.84) (209-140)		4,883,710		
Available for Evaporation		<u>45,277,231</u>		
Evaporation: $45,277,231 \div 1023$	44,259			
Syrup to Pans		Calculated	40,001	60.00
		Required	40,000	60.00

CASE 2

EVAPORATION CONDITIONS



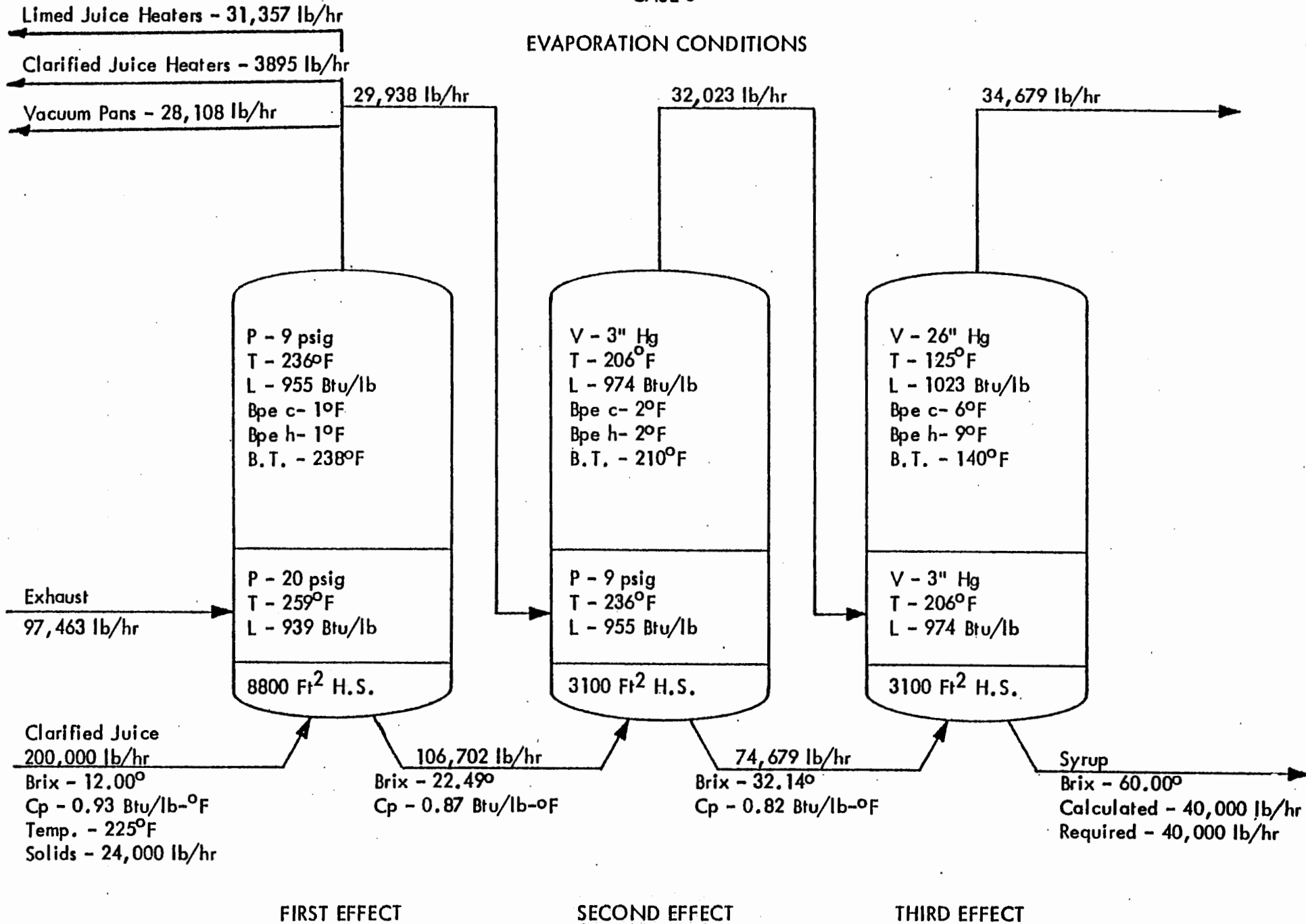
CASE 3

EVAPORATOR MATERIAL AND ENERGY BALANCE

	<u>Exhaust or Vapor, Lb/Hr</u>	<u>Heat Transferred Btu/Hr</u>	<u>Material Lb/Hr</u>	<u>°Brix</u>
<u>First Effect</u>				
Entering: (97,463) (939)	97,463	91,517,757	200,000	12.00
Less: (200,000) (0.93) (238-225)		2,418,000		
Available for Evaporation		<u>39,099,757</u>		
Evaporation: $89,099,757 \div 955$	93,298			
Vapor to Limed Juice Heaters ($29,946,000 \div 955$)	31,357			
Vapor to Clarified Juice Heaters ($3,720,000 \div 955$)	3,895			
Vapor to Vacuum Pans	28,108			
Vapor to Second Effect	29,938			
<u>Second Effect</u>				
Entering: (29,938) (955)	29,938	28,590,790	106,702	22.49
Plus: (106,702) (0.87) (238-210)		2,599,261		
Available for Evaporation		<u>31,190,051</u>		
Evaporation: $31,190,051 \div 974$	32,023			
<u>Third Effect</u>				
Entering:	32,023	31,190,051	74,679	32.14
Plus: (74,679) (0.82) (210-140)		4,286,575		
Available for Evaporation		<u>35,476,626</u>		
Evaporation: $35,476,626 \div 1023$	34,679			
Syrup to Pans		Calculated	40,000	60.00
		Required	40,000	60.00

CASE 3

EVAPORATION CONDITIONS



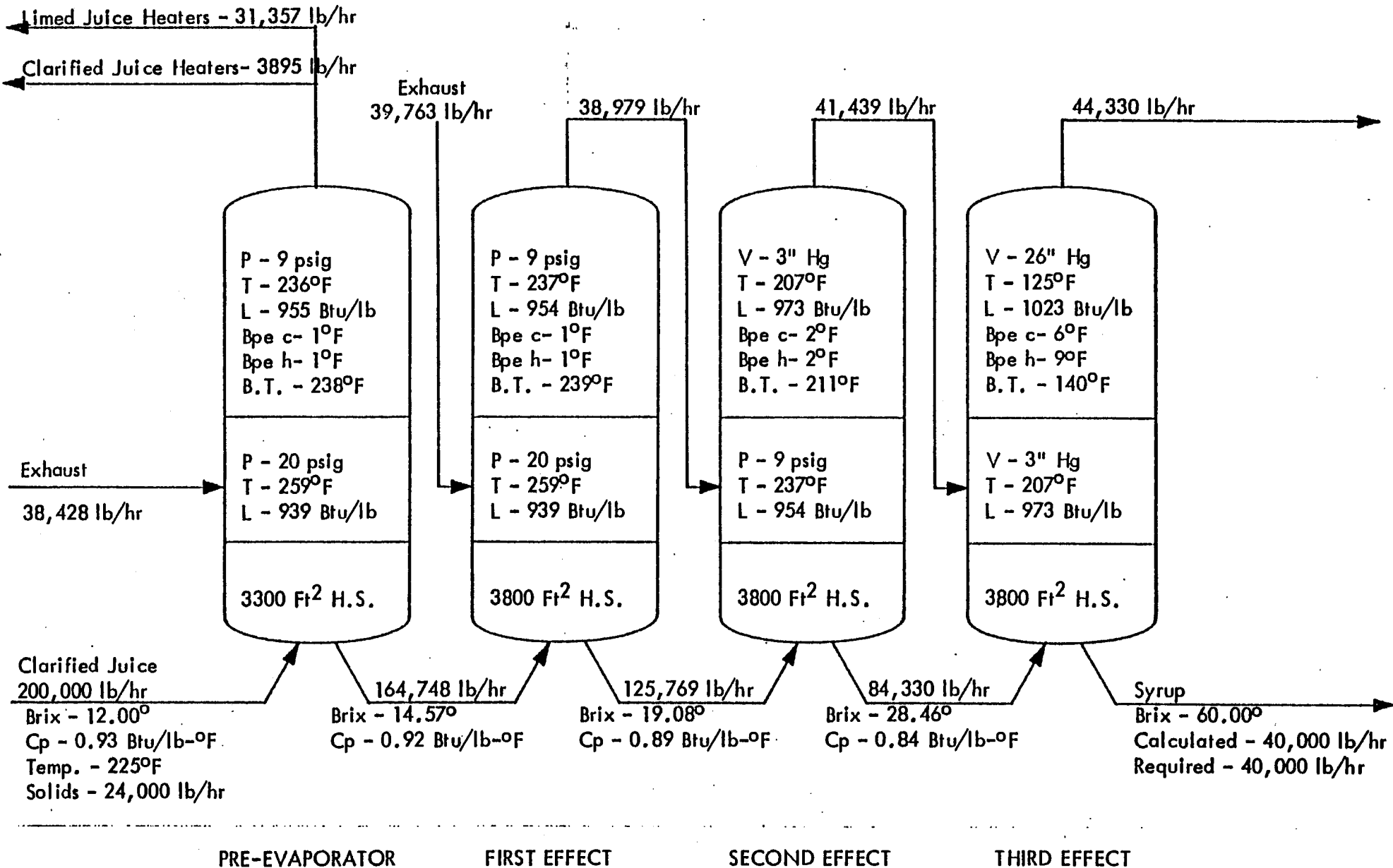
CASE 4

EVAPORATOR MATERIAL AND ENERGY BALANCE

	<u>Exhaust or Vapor, Lb/Hr</u>	<u>Heat Transferred Btu/Hr</u>	<u>Material Lb/Hr</u>	<u>°Brix</u>
<u>Pre-Evaporator</u>				
Entering: (38,428) (939)	38,428	36,083,892	200,000	12.00
Less: (200,000) (0.93) (238-225)		2,418,000		
Available for Evaporation		<u>33,665,892</u>		
Evaporation: $33,665,892 \div 955$	35,252			
Vapor to Limed Juice Heaters ($29,946,000 \div 955$)	31,357			
Vapor to Clarified Juice Heaters ($3,720,000 \div 955$)	3,895			
<u>First Effect</u>				
Entering: (39,763) (939)	39,763	37,337,457	164,748	14.57
Less: (164,748) (0.92) (239-238)		151,568		
Available for Evaporation		<u>37,185,889</u>		
Evaporation: $37,185,889 \div 954$	38,979			
<u>Second Effect</u>				
Entering:	38,979	37,185,889	125,769	19.08
Plus: (125,769) (0.89) (239-211)		3,134,163		
Available for Evaporation		<u>40,320,052</u>		
Evaporation: $40,320,052 \div 973$	41,439			
<u>Third Effect</u>				
Entering:	41,439	40,320,052	84,330	28.46
Plus: (84,330) (0.84) (211-140)		5,029,441		
Available for Evaporation		<u>45,349,493</u>		
Evaporation: $45,349,493 \div 1023$	44,330			
Syrup to Pans		Calculated	40,000	60.00
		Required	40,000	60.00

CASE 4

EVAPORATION CONDITIONS



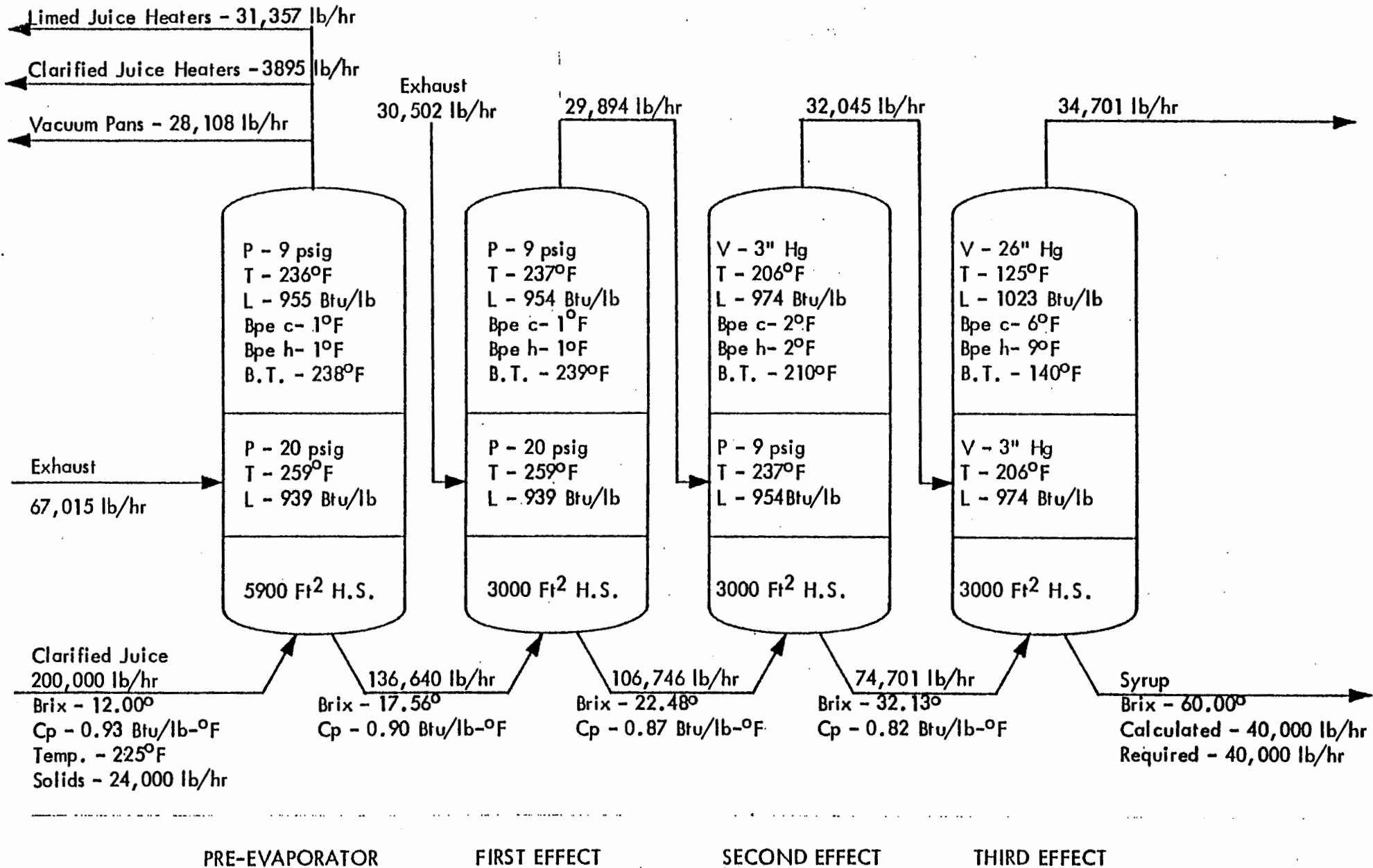
CASE 5

EVAPORATOR MATERIAL AND ENERGY BALANCE

	<u>Exhaust or Vapor, Lb/Hr</u>	<u>Heat Transferred Btu/Hr</u>	<u>Material Lb/Hr</u>	<u>°Brix</u>
<u>Pre-Evaporator</u>				
Entering: (67,015) (939)	67,015	62,927,085	200,000	12.00
Plus: (200,000) (0.93) (238-225)		2,418,000		
Available for Evaporation		<u>60,509,085</u>		
Evaporation: $60,509,085 \div 955$	63,360			
Vapor to Limed Juice Heaters ($29,946,000 \div 955$)	31,357			
Vapor to Clarified Juice Heaters ($3,720,000 \div 955$)	3,895			
Vapor to Vacuum Pans	28,108			
<u>First Effect</u>				
Entering: (30,502) (939)	30,502	28,641,378	136,640	17.56
Less: (136,640) (0.90) (239-238)		122,976		
Available for Evaporation		<u>28,518,402</u>		
Evaporation: $28,518,402 \div 954$	29,894			
<u>Second Effect</u>				
Entering:	29,894	28,518,402	106,746	22.48
Plus: (106,746) (0.87) (239-210)		2,693,202		
Available for Evaporation		<u>31,211,604</u>		
Evaporation: $31,211,604 \div 974$	32,045			
<u>Third Effect</u>				
Entering:	32,045	31,211,604	74,701	32.13
Plus: (74,701) (0.82) (210-140)		4,287,837		
Available for Evaporation		<u>35,499,441</u>		
Evaporation: $35,499,441 \div 1023$	34,701			
Syrup to Pans		Calculated	40,000	60.00
		Required	40,000	60.00

CASE 5

EVAPORATION CONDITIONS



CASE 6

EVAPORATOR MATERIAL AND ENERGY BALANCE

	<u>Exhaust or Vapor, Lb/Hr</u>	<u>Heat Transferred, Btu/Hr</u>	<u>Material Lb/Hr</u>	<u>°Brix</u>
<u>First Effect</u>				
Entering: (40,921)(939)	40,921	38,424,819	200,000	12.00
Less: (200,000)(0.93)(245 - 225)		3,720,000		
Available for Evaporation		<u>34,704,819</u>		
Evaporation: $34,704,819 \div 950$	36,531			
<u>Second Effect</u>				
Entering:	36,531	34,704,819	163,469	14.68
Plus: (163,469)(0.92)(245 - 227)		2,707,047		
Available for Evaporation		<u>37,411,866</u>		
Evaporation: $37,411,866 \div 962$	38,890			
<u>Third Effect</u>				
Entering:	38,890	37,411,866	124,579	19.26
Plus: (124,579)(0.89)(227 - 202)		2,771,883		
Available for Evaporation		<u>40,183,749</u>		
Evaporation: $40,183,749 \div 979$	41,046			
<u>Fourth Effect</u>				
Entering:	41,046	40,183,749	83,533	28.73
Plus: (83,533)(0.84)(202 - 140)		4,350,399		
Available for Evaporation		<u>44,534,148</u>		
Evaporation: $44,534,148 \div 1023$	43,533			
Syrup to Pans		Calculated	40,000	60.00
		Required	40,000	60.00

EVAPORATION CONDITIONS



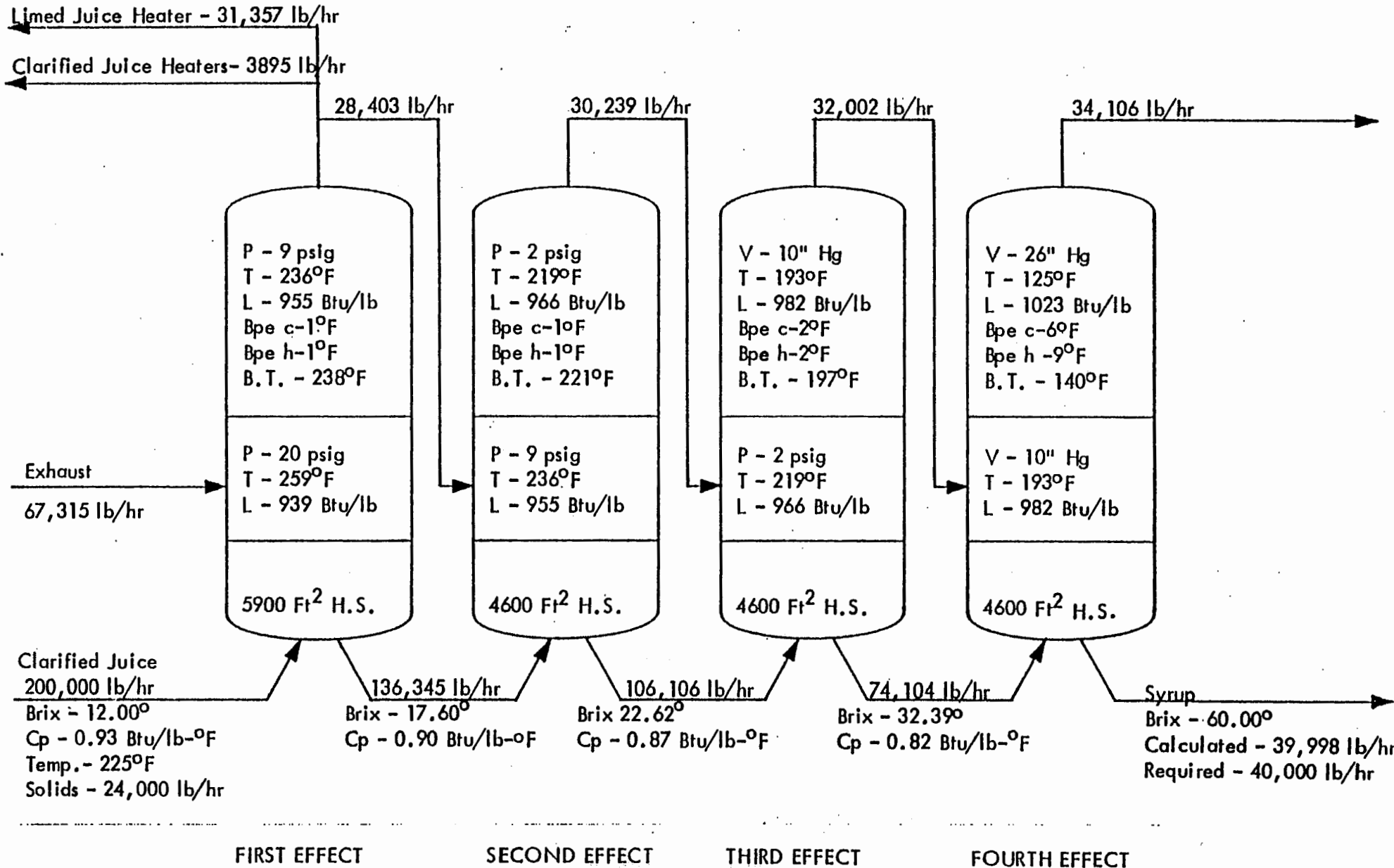
CASE 7

EVAPORATOR MATERIAL AND ENERGY BALANCE

	<u>Exhaust or Vapor, Lb/Hr</u>	<u>Heat Transferred, Btu/Hr</u>	<u>Material Lb/Hr</u>	<u>°Brix</u>
<u>First Effect</u>				
Entering: $(67,315)(939)$	67,315	63,208,785	200,000	12.00
Less: $(200,000)(0.93)(238 - 225)$		2,418,000		
Available for Evaporation		<u>60,790,785</u>		
Evaporation: $60,790,785 \div 955$	63,655			
Vapor to Lined Juice Heaters $(29,946,000 \div 955)$	31,357			
Vapor to Clarified Juice Heater $(3,720,000 \div 955)$	3,895			
Vapor to Second Effect	28,403			
<u>Second Effect</u>				
Entering: $(28,403)(955)$	28,403	27,124,865	136,345	17.60
Plus: $(136,345)(0.90)(238 - 221)$		2,086,079		
Available for Evaporation		<u>29,210,944</u>		
Evaporation: $29,210,944 \div 966$	30,239			
<u>Third Effect</u>				
Entering:	30,239	29,210,944	106,106	22.62
Plus: $(106,106)(0.87)(221 - 197)$		2,215,493		
Available for Evaporation		<u>31,426,437</u>		
Evaporation: $31,426,437 \div 982$	32,002			
<u>Fourth Effect</u>				
Entering:	32,002	31,426,437	74,104	32.39
Plus: $(74,104)(0.82)(197 - 140)$		3,463,621		
Available for Evaporation		<u>34,890,058</u>		
Evaporation: $34,890,058 \div 1023$	34,106			
Syrup to Pans		Calculated	39,998	60.00
		Required	40,000	60.00

CASE 7

EVAPORATION CONDITIONS

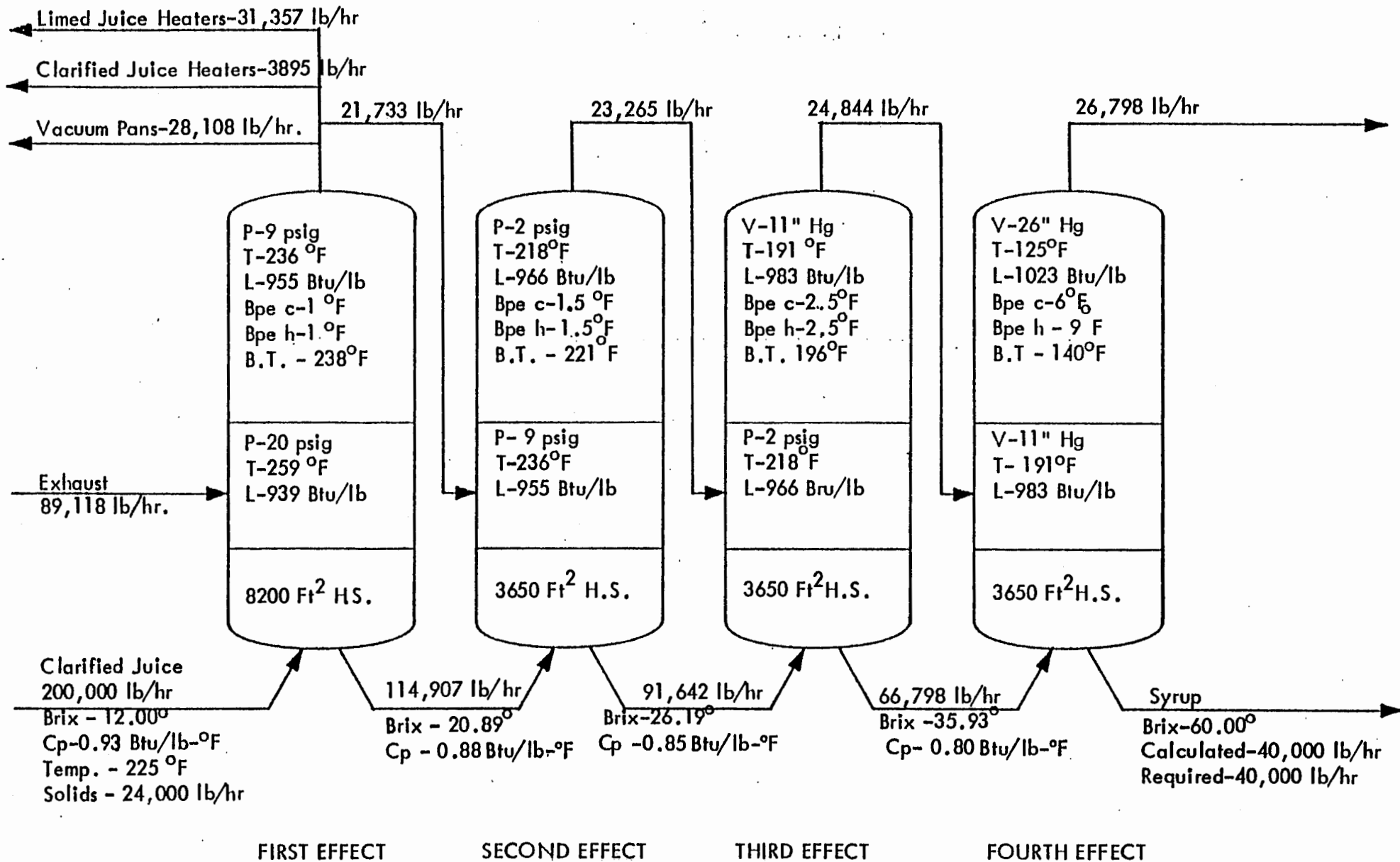


CASE 8

EVAPORATOR MATERIAL AND ENERGY BALANCE

	<u>Exhaust or Vapor, Lb/Hr</u>	<u>Heat Transferred Btu/Hr</u>	<u>Material, Lb/Hr</u>	<u>°Brix</u>
<u>First Effect</u>				
Entering: (89,118)(939)	89,118	83,681,802	200,000	12.00
Less: (200,000)(0.93)(238 - 225)		2,418,000		
Available for Evaporation		<u>81,263,802</u>		
Evaporation: $81,263,802 \div 955$	85,093			
Vapor to Limed Juice Heater (29,946,000 $\div$ 955)	31,357			
Vapor to Clarified Juice Heaters (3,720,000 $\div$ 955)	3,895			
Vapor to Vacuum Pans	28,108			
Vapor to Second Effect	21,733			
<u>Second Effect</u>				
Entering: (21,733)(955)	21,733	20,755,015	114,907	20.89
Plus: (114,907)(0.88)(238 - 221)		1,719,009		
Available for Evaporation		<u>22,474,024</u>		
Evaporation: $22,474,024 \div 966$	23,265			
<u>Third Effect</u>				
Entering:	23,265	22,474,024	91,642	26.19
Plus: (91,642)(0.85)(221 - 196)		1,947,393		
Available for Evaporation		<u>24,421,417</u>		
Evaporation: $24,421,417 \div 983$	24,844			
<u>Fourth Effect</u>				
Entering:	24,844	24,421,417	66,798	35.93
Plus: (66,798)(0.80)(196 - 140)		2,992,550		
Available for Evaporation		<u>27,413,967</u>		
Evaporation: $27,413,967 \div 1023$	26,798			
Syrup to Pans		Calculated	40,000	60.00
		Required	40,000	60.00

CASE 8
EVAPORATION CONDITIONS

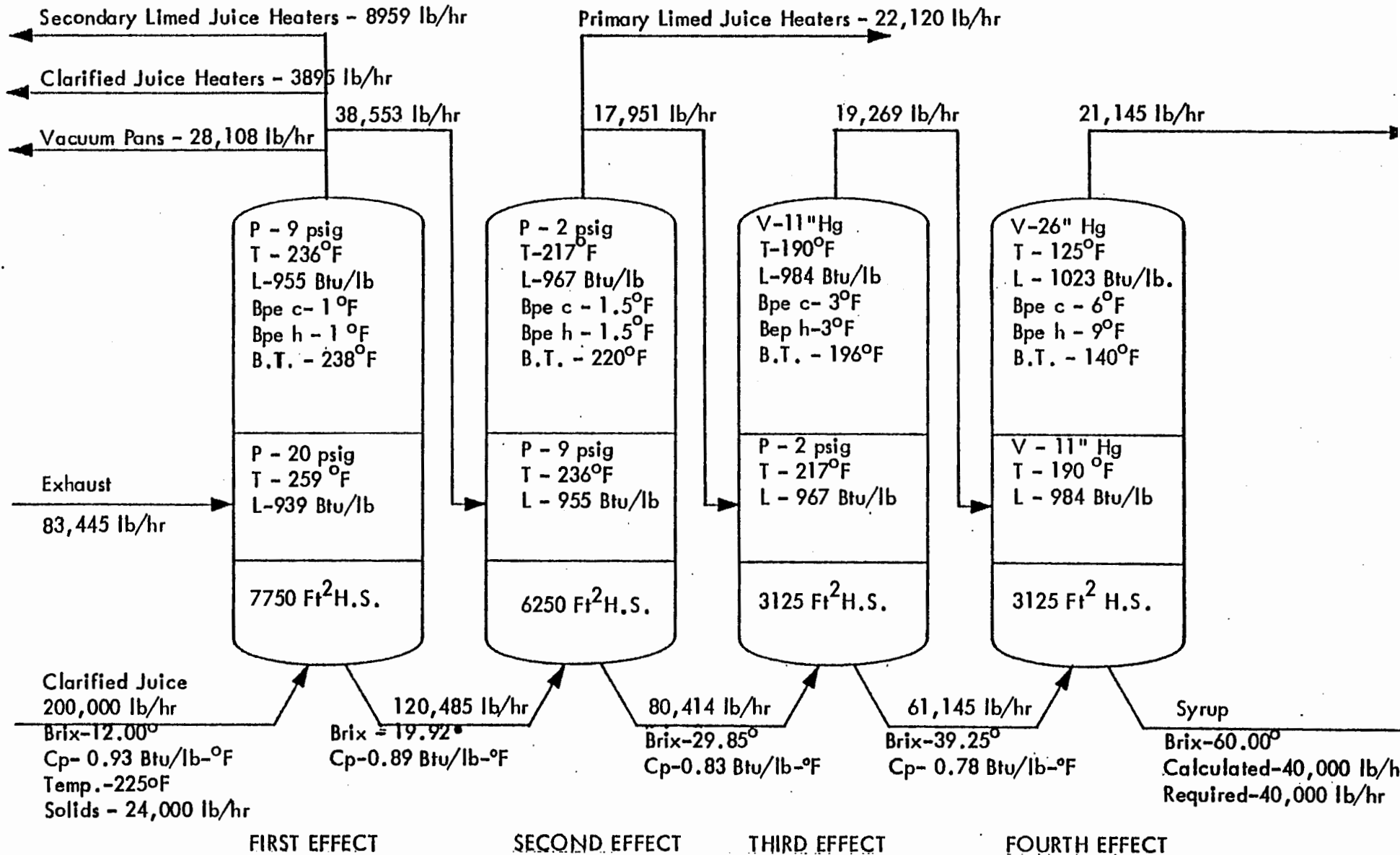


CASE 9

EVAPORATOR MATERIAL AND ENERGY BALANCE

	<u>Exhaust or Vapor, Lb/Hr</u>	<u>Heat Transferred Btu/Hr</u>	<u>Material Lb/Hr</u>	<u>°Brix</u>
First Effect				
Entering: (83,445)(939)	83,445	78,354,855	200,000	12.00
Less: (200,000)(0.93)(238 - 225)		2,418,000		
Available for Evaporation		<u>75,936,855</u>		
Evaporation: $75,936,855 \div 955$	79,515			
Vapor to Secondary Limed Juice Heater. (8,556,000 $\div$ 955)	8,959			
Vapor to Clarified Juice Heaters (3,720,000 $\div$ 955)	3,895			
Vapor to Vacuum Pans	28,108			
Vapor to Second Effect	38,553			
Second Effect				
Entering: (38,553)(955)	38,553	36,818,115	120,485	19.92
Plus: (120,485)(0.89)(238 - 220)		1,930,170		
Available for Evaporation		<u>38,748,285</u>		
Evaporation: $38,748,285 \div 967$	40,071			
Vapor to Primary Juice Heaters (21,390,000 $\div$ 967)	22,120			
Vapor to Third Effect	17,951			
Third Effect				
Entering: (17,951)(967)	17,951	17,358,617	80,414	29.85
Plus: (80,414)(0.83)(220-196)		1,601,847		
Available for Evaporation		<u>18,960,464</u>		
Evaporation: $18,960,464 \div 984$	19,269			
Fourth Effect				
Entering:	19,269	18,960,464	61,145	39.25
Plus: (61,145)(0.78)(196 - 140)		2,670,814		
Available for Evaporation		<u>21,631,278</u>		
Evaporation: $21,631,278 \div 1023$	21,145			
Syrup to Pans				
		Calculated	40,000	60.00
		Required	40,000	60.00

CASE 9 EVAPORATION CONDITIONS

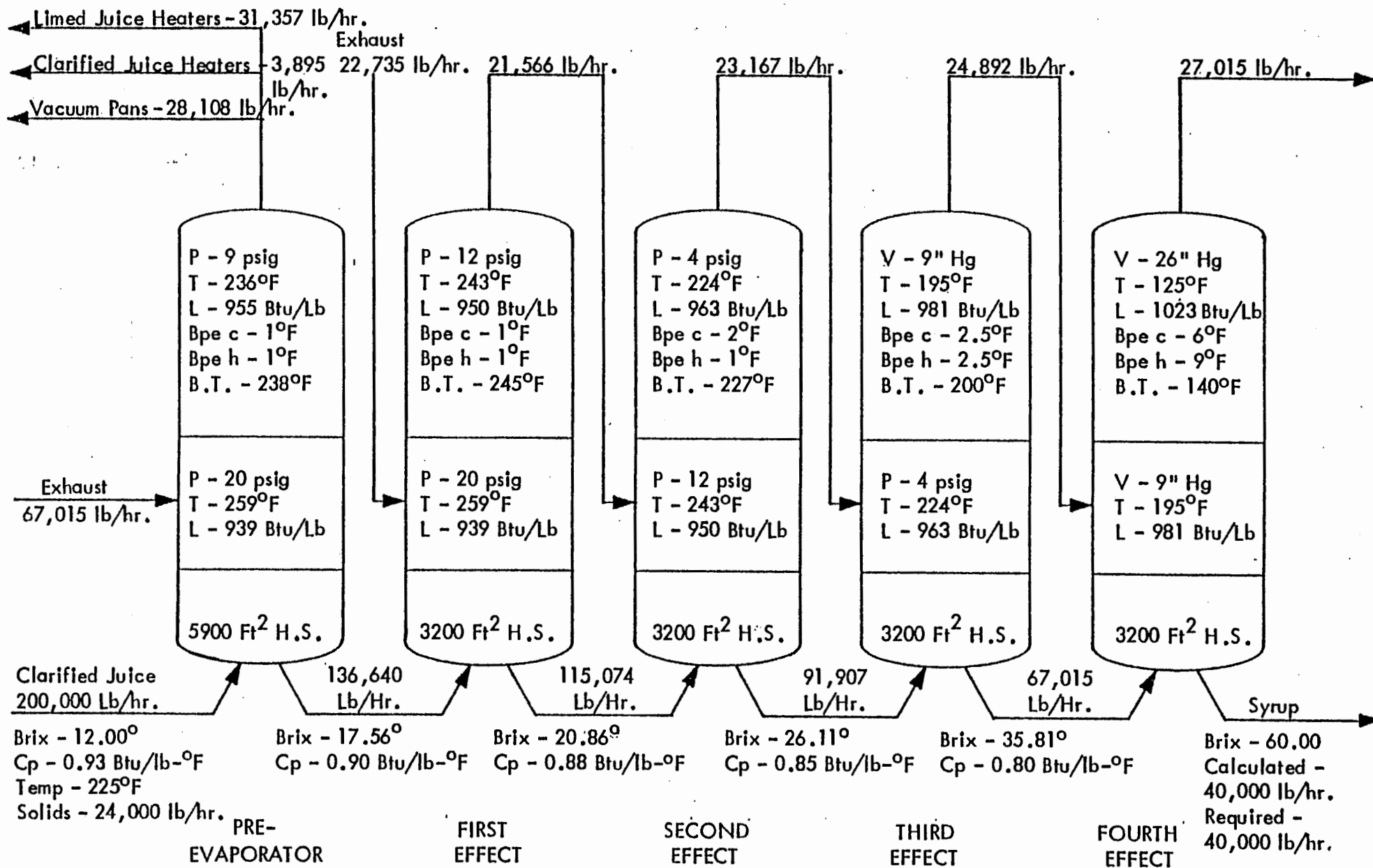


CASE 10

EVAPORATOR MATERIAL AND ENERGY BALANCE

	Exhaust or Vapor, Lb/Hr	Heat Transferred Btu/Hr	Material Lb/Hr	°Brix
<u>Pre-Evaporator</u>				
Entering:				
Less: (200,000) (939)	67,015	62,927,085	200,000	12.00
Available (93) (238-225)		2,418,000		
Evaporator Evaporation		<u>60,509,085</u>		
Vapor to 109,085 ÷ 955	63,360			
Vapor to Juice Heaters (29,946,000 ÷ 955)	31,357			
Vapor to 4 Juice Heaters (3,720,000 ÷ 955)	3,895			
Pans	28,108			
<u>First Effect</u>				
Entering:				
Less: (136,640) (939)	22,735	21,348,165	136,640	17.56
Available (90.90) (245-238)		860,832		
Evaporator Evaporation		<u>20,487,333</u>		
20,487,333 ÷ 950	21,566			
<u>Second Effect</u>				
Entering				
Plus: (115,074)	21,566	20,487,333	115,074	20.86
Available (0.88) (245-227)		1,822,772		
Evaporator Evaporation		<u>22,310,105</u>		
22,310,105 ÷ 963	23,167			
<u>Third Effect</u>				
Entering				
Plus: (91,907)	23,167	22,310,105	91,907	26.11
Available (0.85) (227-200)		2,109,266		
Evaporator Evaporation		<u>24,419,371</u>		
24,419,371 ÷ 981	24,892			
<u>Fourth Effect</u>				
Entering				
Plus: (67,015)	24,892	24,419,371	67,015	35.81
Available (5) (0.80) (200-140)		3,216,720		
Evaporator Evaporation		<u>27,636,091</u>		
27,636,091 ÷ 1023	27,015			
Syrup to				
		Calculated	40,000	60.00
		Required	40,000	60.00

CASE 10 EVAPORATION CONDITIONS

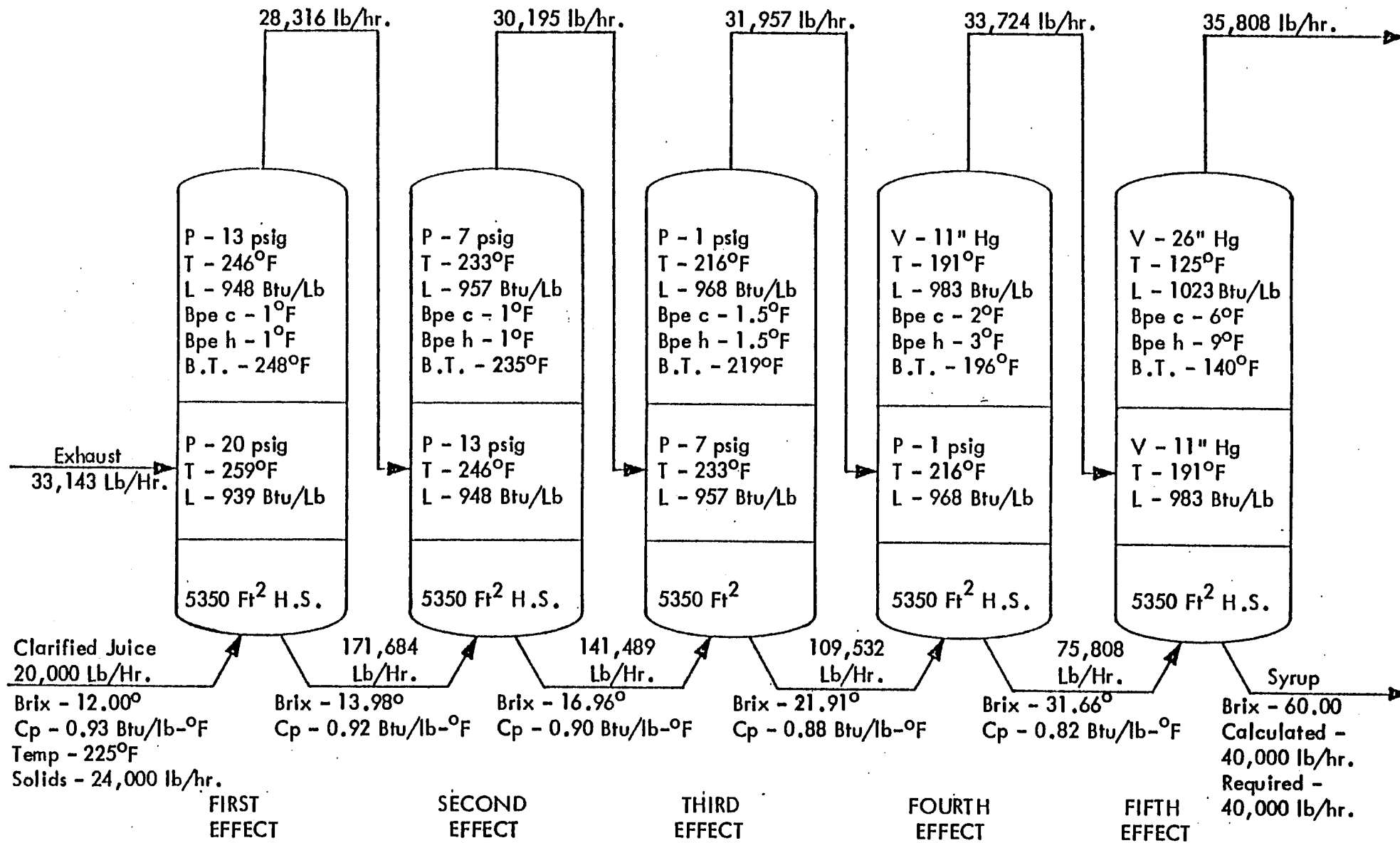


CASE 11

EVAPORATOR MATERIAL AND ENERGY BALANCE

	<u>Exhaust or Vapor, Lb/Hr</u>	<u>Heat Transferred Btu/Hr</u>	<u>Material Lb/Hr</u>	<u>°Brix</u>
<u>First Effect</u>				
Entering: (33,143)(939)	33,143	31,121,277	200,000	12.00
Less: (209,000)(0.93)(248 - 225)		4,278,000		
Available for Evaporation		<u>26,843,277</u>		
Evaporation: $26,843,277 \div 948$	28,316			
<u>Second Effect</u>				
Entering:	28,316	26,843,277	171,684	13.98
Plus: (171,684)(0.92)(248 - 235)		2,053,341		
Available for Evaporation		<u>28,896,618</u>		
Evaporation: $28,896,618 \div 957$	30,195			
<u>Third Effect</u>				
Entering:	30,195	28,896,618	141,489	16.96
Plus: (141,489)(0.90)(235 - 219)		2,037,442		
Available for Evaporation		<u>30,934,060</u>		
Evaporation: $30,934,060 \div 968$	31,957			
<u>Fourth Effect</u>				
Entering:	31,957	30,934,060	109,532	21.91
Plus: (109,532)(0.88)(219 - 196)		2,216,928		
Available for Evaporation		<u>33,150,988</u>		
Evaporation: $33,150,988 \div 983$	33,724			
<u>Fifth Effect</u>				
Entering:	33,724	33,150,988	75,808	31.66
Plus: (75,808)(0.82)(196 - 140)		3,481,103		
Available for Evaporation		<u>36,632,091</u>		
Evaporation: $36,632,091 \div 1023$	35,808			
Syrup to Pans		Calculated	40,000	60.00
		Required	40,000	60.00

CASE 11
EVAPORATION CONDITIONS



CASE 12

EVAPORATOR MATERIAL AND ENERGY BALANCE

	<u>Exhaust or Vapors, Lb/Hr</u>	<u>Heat Transferred Btu/Hr</u>	<u>Material Lb/Hr</u>	<u>° Brix</u>
First Effect				
Entering: $(79,653)(939)$	79,653	74,794,167	200,000	12.00
Less: $(200,000)(0.93)(238 - 225)$		2,418,000		
Available for Evaporation		<u>72,376,167</u>		
Evaporation: $72,376,167 \div 955$	75,787			
Vapor to Secondary Lined Juice Heaters $(8,556,000 \div 955)$	8,959			
Vapor to Clarified Juice Heaters $(3,720,000 \div 955)$	3,895			
Vapor to Vacuum Pans	28,108			
Vapor to Second Effect	34,825			
Second Effect				
Entering: $(34,825)(955)$	34,825	33,257,875	124,213	19.32
Plus: $(124,213)(0.89)(238 - 225)$		1,437,144		
Available for Evaporation		<u>34,695,019</u>		
Evaporation: $34,695,019 \div 964$	35,991			
Vapor to Primary Juice Heaters $(21,390,000 \div 964)$	22,189			
Vapor to Third Effect	13,802			
Third Effect				
Entering: $(13,802)(964)$	13,802	13,305,128	88,222	27.20
Plus: $(88,222)(0.85)(225 - 210)$		1,124,831		
Available for Evaporation		<u>14,429,959</u>		
Evaporation: $14,429,959 \div 974$	14,815			
Fourth Effect				
Entering:	14,815	14,429,959	73,407	32.69
Plus: $(73,407)(0.82)(210 - 188)$		1,324,262		
Available for Evaporation		<u>15,754,221</u>		
Evaporation: $15,754,221 \div 989$	15,929			

Case 12 (continued)

	<u>Exhaust or Vapors, Lb/Hr</u>	<u>Heat Transferred Btu/Hr</u>	<u>Material Lb/Hr</u>	<u>°Brix</u>
<u>Fifth Effect</u>				
Entering:	15,929	15,754,221	57,478	41.76
Plus: $(57,478)(0.77)(188 - 140)$		2,124,387		
Available for Evaporation		<u>17,878,608</u>		
Evaporation: $17,878,608 \div 1023$	17,477			
Syrup to Pans		Calculated Required	40,001 40,000	60.00

EVAPORATION CONDITIONS



CASE 13

EVAPORATOR MATERIAL AND ENERGY BALANCE

	<u>Exhaust or Vapor, Lb/Hr</u>	<u>Heat Transferred, Btu/Hr</u>	<u>Material Lb/Hr</u>	<u>°Brix</u>
First Effect				
Entering: (74,264)(939)	74,264	69,733,896	200,000	12.00
Less: (200,000)(0.93)(246 - 225)		3,906,000		
Available for Evaporation		<u>65,827,896</u>		
Evaporation: $65,827,896 \div 949$	69,366			
Vapor to Clarified Juice Heaters (3,720,000 $\div$ 949)	3,920			
Vapor to Vacuum Pans	28,108			
Vapor to Second Effect	37,338			
Second Effect				
Entering: (37,388)(949)	37,388	35,433,762	130,634	18.37
Plus: (130,634)(0.90)(246 - 232)		1,645,988		
Available for Evaporation		<u>37,079,750</u>		
Evaporation: $37,079,750 \div 959$	38,665			
Vapor to Secondary Limed Juice Heaters (8,556,000 $\div$ 959)	8,922			
Vapor to Third Effect	29,743			
Third Effect				
Entering: (29,743)(959)	29,743	28,523,537	91,969	26.10
Plus: (91,969)(0.85)(232 - 214)		1,407,126		
Available for Evaporation		<u>29,930,663</u>		
Evaporation: $29,930,663 \div 972$	30,793			
Vapor to Primary Limed Juice Heaters (21,390,000 $\div$ 972)	22,006			
Vapor to Fourth Effect	8,787			
Fourth Effect				
Entering: (8,787)(972)	8,787	8,540,964	61,176	39.23
Plus: (61,176)(0.78)(214 - 189)		1,192,932		
Available for Evaporation		<u>9,733,896</u>		
Evaporation : $9,733,896 \div 989$	9,842			

Case 13 (continued)

	<u>Exhaust or Vapor, Lb/Hr</u>	<u>Heat Transferred Btu/Hr</u>	<u>Material Lb/Hr</u>	<u>°Brix</u>
<u>Fifth Effect</u>				
Entering:	9,842	9,733,896	51,334	46.76
Plus: $(51,334)(0.74)(189 - 140)$		1,861,371		
Available for Evaporation		<u>11,595,267</u>		
Evaporation: $11,545,267 \div 1023$	11,335			
Syrup to Pans		Calculated	39,999	60.00
		Required	40,000	60.00

CASE 13

EVAPORATION CONDITIONS

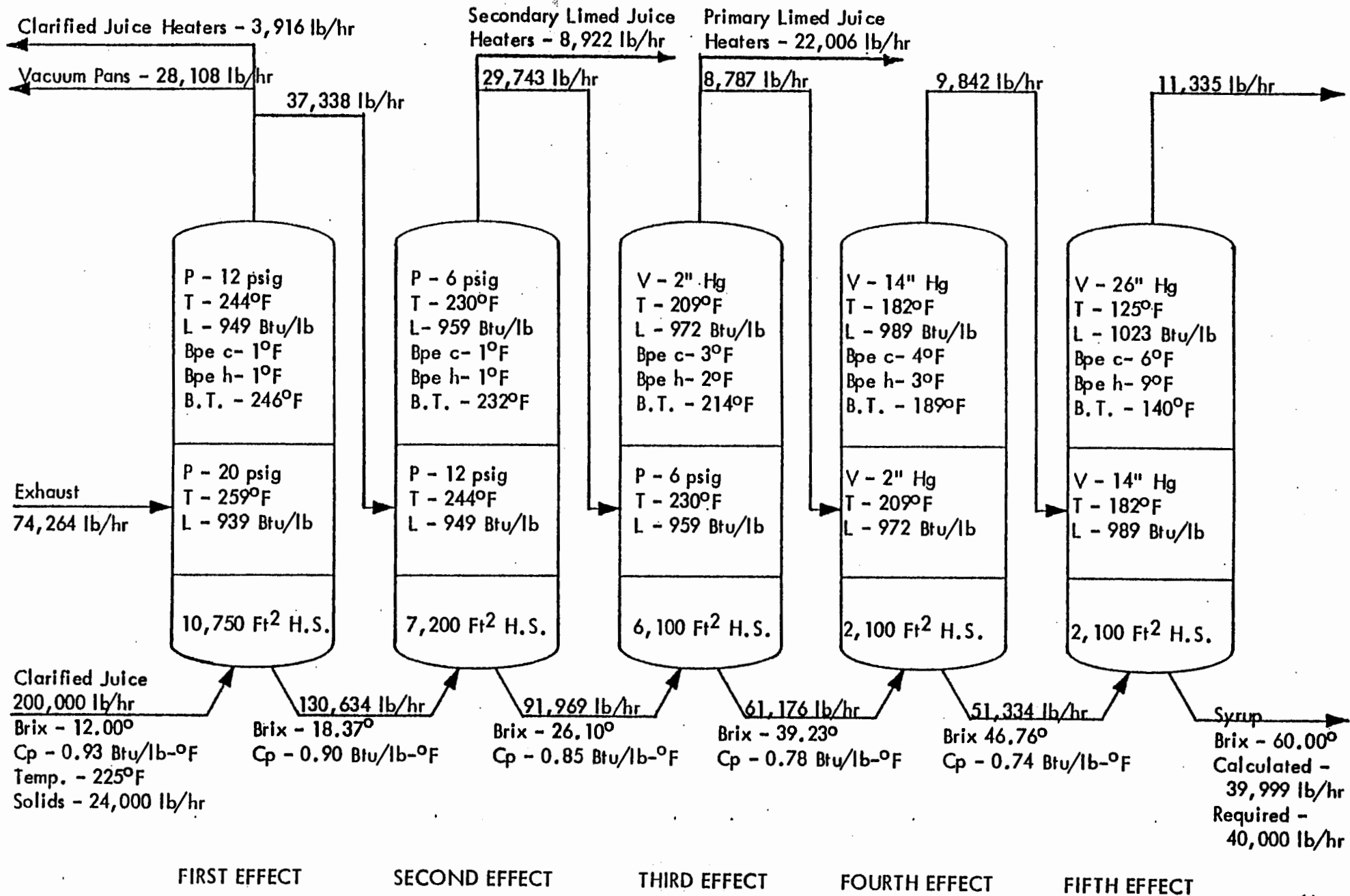


TABLE 1
SUMMARY OF EVAPORATOR CASES
(Basis: 100 Tons Juice Per Hour)

	Triples					Quadruples					Quintuples		
Evaporator Case	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7	Case 8	Case 9	Case 10	Case 11	Case 12	Case 13
<u>Evaporator Heating Surface, Ft²</u>													
Pre-Evaporator	-	-	-	3,300	5,900	-	-	-	-	5,900	-	-	-
First Effect	4,800	6,700	8,800	3,800	3,000	5,100	5,900	8,200	7,750	3,200	5,350	13,700	10,750
Second Effect	4,800	3,900	3,100	3,800	3,000	5,100	4,600	3,650	6,250	3,200	5,350	8,150	7,200
Third Effect	4,800	3,900	3,100	3,800	3,000	5,100	4,600	3,650	3,125	3,200	5,350	3,600	6,100
Fourth Effect	-	-	-	-	-	5,100	4,600	3,650	3,125	3,200	5,350	3,600	2,100
Fifth Effect	-	-	-	-	-	-	-	-	-	-	5,350	3,600	2,100
Total Evaporator Heating Surface, Ft ²	14,400	14,500	15,000	14,700	14,900	20,400	19,700	19,150	20,250	18,700	26,750	32,650	28,250
<u>Heater Heating Surface, Ft²</u>													
Primary Lined Juice Heaters	1,630	2,498	2,435	2,435	2,435	1,630	2,435	2,435	1,400	2,435	1,630	1,303	1,596
Secondary Lined Juice Heater	0	0	0	0	0	0	0	0	1,340	0	0	1,340	1,722
Clarified Juice Heaters	344	818	771	771	771	344	771	771	771	771	344	771	518
Total Heater Heating Surface, Ft ²	1,974	3,316	3,206	3,206	3,206	1,974	3,206	3,206	3,511	3,206	1,974	3,414	3,836
Total Evaporator and Heater Heating Surface, Ft ²	16,374	17,816	18,206	17,906	18,106	22,374	22,906	22,356	23,761	21,906	28,724	36,064	32,086
<u>Exhaust Requirements, Lb/Hr</u>													
Pre-Evaporator	0	0	0	38,428	67,015	0	0	0	0	67,015	0	0	0
Evaporator	53,902	79,910	97,463	39,763	30,502	40,921	67,315	89,118	83,445	22,735	33,143	79,653	74,264
Primary Lined Juice Heaters	31,891	0	0	0	0	31,891	0	0	0	0	31,891	0	0
Secondary Lined Juice Heaters	0	0	0	0	0	0	0	0	0	0	0	0	0
Clarified Juice Heaters	3,962	0	0	0	0	3,962	0	0	0	0	3,962	0	0
Vacuum Pans	28,108	28,108	0	28,108	0	28,108	28,108	0	0	0	28,108	0	0
Total Boiling House Exhaust Requirements	117,863	108,018	97,463	106,299	97,517	104,882	95,423	89,118	83,445	89,750	97,104	79,653	74,264
<u>Evaporator Condenser</u>													
Vapors to be condensed, Lb/Hr	56,413	44,259	34,679	44,330	34,701	43,533	34,106	26,798	21,145	27,015	35,808	17,477	11,335
Injection Water GPM (In 100°F, Out 120°F)	5,789	4,542	3,559	4,549	3,561	4,467	3,500	2,750	2,170	2,772	3,675	1,794	1,163
<u>Relative Heating Surface (Evaporator & Heaters)</u>													
Required	100.0	108.8	111.2	109.4	110.6	136.6	139.9	136.5	145.1	133.8	175.4	220.3	196.0
Relative Exhaust Required	100.0	91.6	82.7	90.2	82.7	89.0	81.0	75.6	70.8	76.1	82.4	67.6	63.0
Relative Evaporator Injection Water Requirement	100.0	78.5	61.5	78.6	61.5	77.2	60.5	47.5	37.5	47.8	63.5	31.0	20.1
Relative Total (Evaporator & Pans) Injection Water Requirement	100.0	84.8	72.8	84.9	72.9	83.9	72.1	63.0	55.9	63.2	74.2	51.3	43.6

TABLE 2

SUMMARY OF LIMED JUICES HEATER CALCULATIONS

(Basis: 100 Tons Juice Per Hour)

	Triples					Quadruples					Quintuples		
Mixed Juice, Tons/Hour	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Mixed Juice, Lb./Hr.	200,000	200,000	200,000	200,000	200,000	200,000	200,000	200,000	200,000	200,000	200,000	200,000	200,000
Filtrates, Lb./Hr. (15% of Mixed Juice)	30,000	30,000	30,000	30,000	30,000	30,000	30,000	30,000	30,000	30,000	30,000	30,000	30,000
Limed Juice, Lb./Hr.	230,000	230,000	230,000	230,000	230,000	230,000	230,000	230,000	230,000	230,000	230,000	230,000	230,000
First Stage Juice Heaters	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7	Case 8	Case 9	Case 10	Case 11	Case 12	Case 13
Juice Temperature, Inlet, °F	80	80	80	80	80	80	80	80	80	80	80	80	80
Juice Temperature, Outlet, °F	220	220	220	220	220	220	220	220	180	220	220	180	180
Juice Temperature Change, °F	140	140	140	140	140	140	140	140	100	140	140	100	100
Specific Heat of Juice at 12° Brix, Btu/lb-°F	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93
Heat Transferred, Btu/Hr.	29,946,000	29,946,000	29,946,000	29,946,000	29,946,000	29,946,000	29,946,000	29,946,000	21,390,000	29,946,000	29,946,000	21,390,000	21,390,000
Pressure of Exhaust or Vapor, psig	20	8	9	9	9	20	9	9	2	9	20	3	2" Hg.
Temperature of Exhaust or Vapor, °F	259	235	236	236	236	259	236	236	217	236	259	222	209
Mean Logarithmic Temperature Difference, °F	91.87	59.95	61.48	61.48	61.48	91.87	61.48	61.48	76.39	61.48	91.87	82.09	67.00
Overall Heat Transfer Coefficient, Btu/Hr-Ft ² -°F	200	200	200	200	200	200	200	200	200	200	200	200	200
Total Heating Surface Required, Ft ²	1,630	2,498	2,435	2,435	2,435	1,630	2,435	2,435	1,400	2,435	1,360	1,303	1,596
Latent Heat of Exhaust or Vapors, Btu/lb.	939	955	955	955	955	939	955	955	967	955	939	964	972
Exhaust or Vapors Required for First Stage Juice Heating, Lb./Hr.	31,891	31,357	31,357	31,357	31,357	31,891	31,357	31,357	22,120	31,357	31,891	22,189	22,006
Second Stage Juice Heaters													
Juice Temperature Into Second Stage, °F	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	180	NONE	NONE	180	180
Juice Temperature Out of Second Stage, °F									220			220	220
Juice Temperature Change, °F									40			40	40
Specific Heat of Juice at 12° Brix, Btu/lb-°F									0.93			0.93	0.93
Heat Transferred, Btu/Hr.									8,556,000			8,556,000	8,556,000
Pressure of Exhaust or Vapor, psig									9			9	6
Temperature of Exhaust or Vapor, °F									236			236	230
Mean Logarithmic Temperature Difference, °F									31.93			31.93	24.85
Overall Heat Transfer Coefficient, Btu/Hr-Ft ² -°F									200			200	200
Total Heating Surface Required, Ft ²									1,340			1,340	1,722
Latent Heat of Exhaust or Vapors, Btu/lb.									955			955	959
Exhaust or Vapors Required for Second Stage Juice Heating, Lb./Hr.									8,959			8,959	8,922

TABLE 3
SUMMARY OF CLARIFIED JUICE HEATER CALCULATIONS
(Basis: 100 Tons Juice Per Hour)

	Triples					Quadruples					Quintuples		
Clarified Juice, Tons/Hour	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Clarified Juice, Lbs/Hour	200,000	200,000	200,000	200,000	200,000	200,000	200,000	200,000	200,000	200,000	200,000	200,000	200,000
Clarified Juice Heating	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7	Case 8	Case 9	Case 10	Case 11	Case 12	Case 13
Juice Temperature, Inlet, °F	205	205	205	205	205	205	205	205	205	205	205	205	205
Juice Temperature, Outlet, °F	225	225	225	225	225	225	225	225	225	225	225	225	225
Juice Temperature Change, °F	20	20	20	20	20	20	20	20	20	20	20	20	20
Specific Heat of Juice at 12° Brix, Btu/lb-°F	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93
Heat Transferred, Btu/ltr.	3,720,000	3,720,000	3,720,000	3,720,000	3,720,000	3,720,000	3,720,000	3,720,000	3,720,000	3,720,000	3,720,000	3,720,000	3,720,000
Pressure of Exhaust or Vapors, psig	20	8	9	9	9	20	9	9	9	9	20	9	12
Temperature of Exhaust or Vapor, °F	259	235	236	236	236	259	236	236	236	236	259	236	244
Mean Logarithmic Temperature Difference, °F	43.23	18.20	19.30	19.30	19.30	43.23	19.30	19.30	19.30	19.30	43.23	19.30	28.71
Overall Heat Transfer Coefficient, Btu/ltr-Ft ² -°F	250	250	250	250	250	250	250	250	250	250	250	250	250
Total Heating Surface Required, Ft ²	344	818	771	771	771	344	771	771	771	771	344	771	518
Latent Heat of Exhaust or Vapors, Btu/lb.	939	955	955	955	955	939	955	955	955	955	939	955	949
Exhaust or Vapors Required for Clarified Juice Heating, Lb/ltr.	3,962	3,895	3,895	3,895	3,895	3,962	3,895	3,895	3,895	3,895	3,962	3,985	3,920

TABLE 4

SUMMARY OF HEAT TRANSFER COEFFICIENTS AND EVAPORATION RATES

(Basis: 100 Tons Juice Per Hour)

	Triples					Quadruples					Quintuples		
Evaporator Case No.	1	2	3	4	5	6	7	8	9	10	11	12	13
Evaporation Rate, Lb/hr-Ft ² of H.S.													
Pre-Evaporator	-	-	-	10.68	10.74	-	-	-	-	10.74	-	-	-
First Effect	10.48	11.08	10.60	10.26	9.96	7.16	10.79	10.38	10.26	6.74	5.29	5.53	6.45
Second Effect	11.10	10.63	10.33	10.91	10.68	7.63	6.57	6.37	6.41	7.24	5.64	4.42	5.37
Third Effect	11.75	11.35	11.19	11.67	11.57	8.05	6.96	6.81	6.17	7.78	5.97	4.12	5.05
Fourth Effect	-	-	-	-	-	8.54	7.41	7.34	6.77	8.44	6.30	4.42	4.69
Fifth Effect	-	-	-	-	-	-	-	-	-	-	6.69	4.85	5.40
Overall	11.11	11.03	10.67	10.88	10.74	7.84	8.12	8.36	7.90	8.56	5.98	4.90	5.65
Overall Heat Transfer Coefficient, Btu/Hr-Ft ² -°F													
Pre-Evaporator	-	-	-	521	508	-	-	-	-	508	-	-	-
First Effect	527	496	495	491	477	538	510	486	481	477	529	496	499
Second Effect	385	367	355	376	352	425	393	379	368	400	456	371	410
Third Effect	159	157	152	158	158	319	289	280	265	290	386	308	292
Fourth Effect	-	-	-	-	-	136	129	131	121	139	289	223	203
Fifth Effect	-	-	-	-	-	-	-	-	-	-	122	104	110
Design (Dessin) Overall Heat Transfer Coefficient, Btu/Hr-Ft ² -°F.													
Pre-Evaporator	-	-	-	526	508	-	-	-	-	508	-	-	-
First Effect	517	498	477	498	477	523	508	487	493	485	526	497	500
Second Effect	391	366	350	372	354	439	396	378	360	402	461	372	404
Third Effect	160	155	155	158	155	331	295	277	260	296	389	302	295
Fourth Effect	-	-	-	-	-	139	129	125	123	133	289	219	208
Fifth Effect	-	-	-	-	-	-	-	-	-	-	125	106	106

ACKNOWLEDGEMENT

The author extends his thanks and appreciation to his colleagues at F. C. Schaffer & Associates, Inc. for their many helpful suggestions, and to F. C. Schaffer & Associates, Inc. for permission to publish these results.

REFERENCES

1. Birkett, H. S. 1977. "The Fuel and Steam Balance in the Raw Sugar Factory"
Proc. ASSCT 6 (NS) : 165 - 176.
2. Hugot, E. 1972. Handbook of Cane Sugar Engineering
Second Edition. Elsevier Publishing Company, New York
3. Meade, G.P. 1964. Spencer-Meade Cane Sugar Handbook
Ninth Edition. John Wiley and Sons, Inc., New York